

Durga Nagendra Raghava Kumar Modhugu

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SUMMARY

Machine learning engineer working in computer vision, 3D perception, and on-device inference. My work has spanned academic research (contributing to Ego4D at CVPR and T-PAMI) and industry roles focused on taking models into production. Along the way I have had the chance to help build ML teams, mentor early-career engineers, and collaborate with academic partners including IITs, CMU, UMich, and IISc. I enjoy working on practical problems where research ideas meet real-world constraints.

EXPERIENCE

Track3D

Oct 2025 – Present

Principal Engineer, Machine Learning

- Working with the founding team on the computer vision roadmap for an industrial site monitoring platform, covering data ingestion, model development, and deployment.
- Worked on metric-scale localisation in construction environments using monocular SfM, a long-standing problem for the team; the approach has helped the platform take on more projects.
- Building a perception pipeline that brings together localisation, 3D mapping, and object detection, designed to cope with dust, occlusion, and changing light.
- Helping grow the ML team from 8 to 16 engineers through hiring, interviews, and mentoring.

Qualcomm India Pvt. Limited

Sep 2024 – Oct 2025

Staff Engineer, Machine Learning

- Led a small team working on inference optimisation for Qualcomm AI Hub and automotive customer models.
- Redesigned the quantiser and led its implementation, reducing latency on autonomous-driving CV models by about 40% and performing well on competitive benchmarks.
- Optimised vision and language models (Llama, LVMs) for on-device deployment.
- Added ONNX contrib-op support to the compiler platform and contributed improvements to the quantisation SDK.

Valeo India Pvt. Limited

Sep 2023 – Sep 2024

Senior Lead Engineer, Machine Learning

- Worked on sensor fusion combining RGB cameras and multi-echo LiDAR to build probabilistic occupancy maps in Bird's Eye View (BEV).
- Developed obstacle detection and moving-object detection and segmentation methods that hold up under sensor noise, occlusion, dust, and glare.
- Built camera-soiling localisation to help perception handle degraded camera input.
- Used knowledge distillation and LiDAR data compression to run perception models in real time on embedded automotive hardware.
- Co-authored work on in-cabin driver attention monitoring (IV 2024, ICASSP 2024).
- Co-filed 2 patents: instance segmentation on IR imagery via depth estimation, and RGB-D and LiDAR fusion for point-cloud semantic segmentation using GNNs.

TalentSprint India Pvt. Limited

Apr 2018 – Sep 2023

Director, Machine Learning

- Helped build the ML team, which grew from 3 to 15 engineers over 5 years; enjoyed mentoring graduate hires, several of whom went on to lead ML work.
- Set technical direction for ML across the company's products, working closely with business and product teams.

RESEARCH EXPERIENCE

IIIT Hyderabad

Jan 2019 – Dec 2021

Research Scholar, CSE (GPA: 9.0/10)

- Supervised by Prof. C.V. Jawahar (IIIT Hyderabad) and Prof. Manmohan Chandraker (UC San Diego).
- Worked on parametric scene parsing that combines ground-level and aerial imagery for navigation (ICRA 2021).
- Contributed to Ego4D (CVPR 2022, T-PAMI), a large egocentric video dataset created by a global consortium including Meta AI, MIT, CMU, and NUS.

EDUCATION

MS by Research, Computer Science & Engineering, IIIT Hyderabad

B.E., Electronics & Communication Engineering, Osmania University

SELECTED PUBLICATIONS

- Ego4D: Around the World in 3,000 Hours of Egocentric Video — *T-PAMI 2024; CVPR 2022*
- Looking Farther in Parametric Scene Parsing with Ground and Aerial Imagery — *ICRA 2021*
- Holistic Driver Monitoring: A Multi-Task Approach for In-cabin Driver Attention Evaluation — *IV 2024*
- DriveSafe: Pose-based Driver Monitoring in Software Defined Vehicle Using STGCN — *ICASSP 2024*
- Dear Commissioner, Please Fix These: A Scalable System for Inspecting Road Infrastructure — *NCVPRIPG 2019*

SKILLS

Programming Python, Java, SQL, C, C++

ML & DL PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, NumPy, Pandas

Domains Computer Vision, 3D Vision, SLAM, Localisation, Autonomous Navigation, Egocentric Vision, NLP, On-device Inference, MLOps

TALKS & ACTIVITIES

- Delivered lectures on PyTorch and deep learning architectures (BERT, benchmarking DNNs) to industry professionals.
- Delivered a talk on AI/ML applications in agriculture at NAARM (National Academy of Agricultural Research Management).
- Member of evaluation committee for AI/ML-based defect detection in tile manufacturing pipelines.